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MILITARY SPECIFICATION

TRUCK, FIRE FIGHTING A/S32P-2, AIRCRAFT, WATER AND FOAM, 2,500 GALLON CAPACITY

1. SCOPE

1.1 This specification covers one type of aircraft fire-fighting truck, designated A/S32P-2, of sufficient agent and discharge capacity to control fires incident to large size aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-4699	Truck, Fire Fighting and Rescue, Fog and Foam, General Requirements for
MIL-T-9107	Test Reports, Preparation of
MIL-M-26512	Maintainability Requirements for Aerospace Systems and Equipment
MIL-I-26600	Interference Control Requirements, Aeronautical Equipment
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-808	Finishes, Protective, and Codes, For Finishing Schemes for Ground and Ground Support Equipment

FSC 4210

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DRAWING

FWD Corporation

531000 A/S32P-2 Fire-Fighting Truck

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 General requirements.- The requirements of Drawing Index 531000 and details thereto (see 6.2) apply as requirements of this specification with the exceptions and additions specified herein.

3.3 Design and construction.- The truck shall be designed and constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, and service.

3.3.1 Reliability.- The truck shall have a minimum MTBF (mean time between failure) of 55 hours with a minimum reliability of .86 for an 8-hour mission at a confidence factor of .90.

3.3.2 Maintainability.- Maintainability requirements shall be in accordance with MIL-M-26512.

3.4 Performance

3.4.1 Mobility.- The fully-loaded and equipped truck shall be capable of meeting the following requirements:

a. Maintain a minimum speed of 60 mph on level improved roads

b. Accelerate from a complete stop to a speed of 55 mph in 60 seconds on dry, level, concrete roads

c. Ascend smooth, dry, concrete roads having an 8-percent grade at a maintained speed of 20 mph

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d. Maintain a speed not in excess of 2-1/2 mph without rough, irregular engine operation or engine stalling

e. Upon application of service brakes, stop in a distance of 30 feet from a speed of 20 mph on dry, hard surface, level roads free of loose material.

3.4.2 Immediate response.- The truck shall be capable of immediate response from standby status, including functional operation of component equipment, to any emergency under conditions specified herein. The truck shall be capable of discharging the extinguishing agents immediately upon arrival at the scene of an emergency. The truck shall be capable of maneuvering at low speeds over cross-country terrain (see 6.3.3) normally traversed by wheeled tactical vehicles, with fire-fighting equipment operating at full discharge.

3.4.3 Fire-fighting equipment

3.4.3.1 Fire pump.- The fire (water) pump shall be a single-stage, centrifugal type with a capacity of 1,400 gpm at 250 psi.

3.4.3.2 Foam pump.- The foam pump shall be a single-stage, centrifugal type with a capacity of 125 gpm at 250 psi.

3.4.3.3 Foam proportioners.- The foam proportioners shall be of the eductor type, and shall maintain the proper ratio (ranging from 2-1/2 to 8 percent) of foam concentrate to water under all operating conditions.

3.4.3.4 Nozzles.- When discharging 6-percent foam solution in a solid stream pattern, all nozzles shall be capable of producing foam of not less than 9 expansion (volume of foam solution equal to 9 times the original volumetric liquid discharge of water and foam concentrate) and having a 25-percent drainage time (time for 25-percent of the originally contained liquid to drain from solution) of not less than 6 minutes. When discharging 6-percent foam solution in a widely dispersed pattern, a maximum drop-off of 10-percent below the required solid stream minimum expansion will be permitted.

3.4.3.4.1 Roof turret nozzle.- The roof turret nozzle shall be of the double-barreled, remotely-controlled, power-assist type. The nozzle shall be capable of discharging either foam or water selectively from one or both barrels. The capacity shall be 500 gpm (water rate) through one barrel or 1,000 gpm through both barrels. When discharging

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foam solution at rated capacity, the range of discharge, from nozzle tip to the far side of discharge pattern, shall be at least 70 feet for widely dispersed foam and 150 feet for solid stream foam. The turret shall be manually controllable for all operations in the event of power failure. Manual control from the cab roof will not be required.

3.4.3.4.2 Bumper turret nozzle.- The bumper turret nozzle shall be of the single-barrel, remote, manually-controlled type. The capacity shall be 300 gpm (water rate). When discharging foam solution at rated capacity, the range of discharge, from nozzle tip to the far side of discharge pattern, shall be at least 50 feet for widely dispersed foam and 80 feet for solid stream foam.

3.4.3.4.3 Handline nozzle.- The handline nozzle shall be of the combination foam and water type. The capacity shall be 100 gpm (water rate). When discharging foam solution at rated capacity, the range of discharge, from nozzle tip to the far side of discharge pattern, shall be at least 25 feet for widely dispersed foam and 40 feet for solid stream foam.

3.4.3.4.4 Under-truck nozzles.- Under-truck nozzles shall not be provided.

3.4.3.5 Agent tanks.- The agent tank discharge system shall be such that after discharge through the fire-fighting equipment, the amount of agent (foam concentrate and water) remaining in the agent system will not exceed 3 percent by weight of the initial amount of agent.

3.4.3.5.1 Water tank.- The water tank shall have a 2,300-gallon capacity.

3.4.3.5.2 Foam tank.- The foam tank shall have a 200-gallon capacity.

3.5 Environmental equipment.- The following environmental equipment shall be provided as specified herein and shall conform to the applicable detail drawings:

3.5.1 Winterization kits.- Unless otherwise specified, a class I winterization kit shall be provided on the truck as standard equipment. When specified (see 6.2), a class II winterization kit shall be provided.

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3.5.2 Cab air-conditioning kit.- When specified (see 6.2), an air-conditioning kit shall be installed. The kit shall lower and maintain an inside cab temperature of +90° F or lower, in less than 15 minutes, when the outside ambient temperature is 125° F.

3.6 Power train

3.6.1 Transfer case.- The transfer case shall contain low- and high-range gears to meet tractive effort demands of all conditions of service. The gears shall be of the constant-mesh-clutch-engaged or synchronized-shift design so that shifting from low to high range gears, or vice versa, may be accomplished while the truck is traveling at speeds up to 15 mph. The torque capacities shall be adequate to transmit the maximum torque from the engines through all gear reductions involved. Gears shall be fabricated so as to deliver the service intended, and to operate quietly.

3.7 Part numbering of interchangeable parts.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

3.8 Electromagnetic interference.- The truck shall comply with the requirements of MIL-I-26600.

3.9 Weight.- The gross vehicle weight of the truck, minus the cab air-conditioning kit, shall not exceed 67,000 pounds.

3.10 Finishes and protective coatings.- The truck shall be treated, painted, and marked in accordance with the requirements of MIL-STD-808.

3.11 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.12 Workmanship.- The truck, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws and bolts; et cetera.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests.- The inspection and testing of the truck shall be classified as follows:

a. Preproduction testing. . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Atmospheric conditions.- The performance abilities of the truck, when affected by such conditions and unless otherwise specified, shall be demonstrated at, or corrected for, standard sea-level atmospheric conditions of 29.92 inches Hg barometric pressure, 0.39 inch Hg water vapor pressure, and 60° F ambient air temperature.

4.4 Preproduction testing

4.4.1 Test sample.- The preproduction test sample shall consist of a model representative of the production item.

4.4.2 Test report.- Upon completion of the preproduction tests, a test report shall be prepared in accordance with MIL-T-9107 and one reproducible and two copies of the complete report shall be furnished to the procuring activity.

4.4.2.1 Reliability and maintainability information.- The following information shall be submitted as an attachment accompanying the above report (same quantity and type of copies) or shall be included as part of that report:

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, miles, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded

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b. Description of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified

c. Description of the engineering reasoning verifying corrections made, to be made on production items, or proposed to be made, and the predicted effectiveness of those corrections

d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item

e. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item

f. Estimated (±5 percent of actual experienced) manhours required for each maintenance and servicing action during the tests. A brief description of the qualifications and experience of the personnel involved shall be included and shall be adequate to allow comparison to the personnel anticipated in similar field work.

4.4.3 Preproduction tests.- The preproduction tests shall consist of all tests specified under 4.6, except the following:

a. Five-mile road test. . . . . See 4.6.3

b. Fire-fighting equipment operational check. . . . . See 4.6.4

c. One-hundred-mile road test . . . . . See 4.6.5.6.

4.5 Acceptance tests.- The acceptance tests shall consist of the following:

a. Individual tests . . . . . See 4.5.1

b. Periodic sampling plan and tests . . . . . See 4.5.2

4.5.1 Individual tests.- Each truck shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Mechanical inspection. . . . . See 4.6.2

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c. Five-mile road test. . . . . See 4.6.3

d. Fire-fighting equipment operational check. . . . . See 4.6.4.

4.5.2 Sampling plan and tests

4.5.2.1 Periodic sampling.- One truck shall be selected at random from every 50 or fraction thereof produced and subjected to the following test:

a. Acceleration . . . . . See 4.6.5.2

b. Grade speed. . . . . See 4.6.5.3

c. Minimum speed. . . . . See 4.6.5.5

d. One-hundred-mile road test . . . . . See 4.6.5.6

e. Fire-fighting equipment tests. . . . . See 4.6.7.

4.5.2.1.1 Rejection and retest.- When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.1.2 Individual tests may continue.- For operational reasons, individual tests or sampling plans may be continued pending the investigation of a sampling test failure. Final acceptance of the items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product.- The truck shall be inspected to determine compliance with the requirements of this specification. All features and components shall be inspected with respect to materials, workmanship, dimensions, proper installation, and marking.

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4.6.2 Mechanical inspection.- A critical mechanical inspection shall be made of all components, and data recorded as to condition, defects of manufacture, damage in transit, and damage through use prior to test. All components shall be moved through their maximum normal displacement and shall exhibit no malfunctioning, leaks, objectional rough operation, or irregular movements. The inspection shall include complete lubrication and servicing of the truck in order to insure readiness for operation if required.

4.6.3 Five-mile road test.- The truck shall be road tested for a distance of at least 5 miles, during which time the truck shall be accelerated from a complete stop to its maximum speed, operated in reverse gear, stopped by fast braking, and manipulated through sharp left and right turns.

4.6.4 Fire-fighting equipment operational check.- The agent dispensing system shall be checked by discharging agent, including foam, in the widely dispersed and solid stream patterns through all nozzles while operating at their maximum discharge rates.

4.6.5 Vehicular performance.- The truck shall be loaded to its gross weight, including agent and fire-fighting equipment, and operated over Aberdeen Proving Ground test courses to determine its vehicular performance characteristics as covered by 4.6.5.1 through 4.6.5.5.

4.6.5.1 Endurance test.- The truck shall be operated over the following test courses for the distances specified herein.

a. Improved gravel - 500 miles

b. Belgian block - 50 miles

c. Level cross-country - 375 miles

d. Hilly cross-country - 250 miles

e. Paved roads - 1,000 miles.

(Note: Paved roads traveled by the truck while commuting between the manufacturer's facility and test facility will be considered to satisfy in whole or in part, according to the number of miles traveled, the requirements of the paved roads test.)

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4.6.5.2 Acceleration.- The truck shall be accelerated five times from a complete stop to a speed of 55 mph in 60 seconds on a dry, level, concrete road.

4.6.5.3 Grade speed.- The truck shall be driven up an 8-percent grade to determine sustained grade-speed performance. If an 8-percent grade is not available, actual performance data on three different grades will be determined, and calculations made to determine performance on an 8-percent grade. The speedometer shall be calibrated or a fifth wheel used to determine the true road speeds. The length of the grades shall be recorded.

4.6.5.4 Braking.- The truck shall be driven at a speed of 20 mph on level concrete road, and the distance required to bring the truck to a stop through application of the service brakes shall not exceed 30 feet. The speedometer shall be calibrated or a fifth wheel used to determine the true road speed. The stopping distance shall be measured by using appropriate methods that will eliminate error due to the reaction time of the driver.

4.6.5.5 Minimum speed.- The truck shall be driven at speeds not in excess of 2-1/2 mph on level concrete road and the operation observed. There shall be no evidence of rough, irregular engine operation or engine stalling.

4.6.5.6 One-hundred-mile road test.- The truck shall be operated over improved roads for a distance of 5 miles at a minimum speed of 60 mph and for a distance of 95 miles at an average speed of 45 mph. During this test, the following functions shall be performed and the truck shall be examined for malfunctions:

a. Normal or average stops from a speed of at least 45 mph - 10 stops

b. Quick stops, from a speed of at least 20 mph, by instantaneous application of full service braking power - 5 stops

c. Sharp (maximum cramping angle) right turns and sharp left turns from a speed of at least 10 mph - 5 turns each.

4.6.6 Electromagnetic interference.- The truck shall be tested for radiated interference in accordance with MIL-I-26600 for class Ib equipment. The antenna for this test shall be placed 6 feet from the outer perimeter at the point of maximum interference.

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4.6.7 Fire-fighting equipment.- The following tests shall be conducted on the fire-fighting equipment to determine compliance with the requirements of this specification.

4.6.7.1 Agent tank exhaustion.- The agent tanks shall be filled with foam concentrate and water, and discharge made through all nozzles. At the completion of effective discharge, the agent remaining in the system (agent tanks and lines) shall be drained and weighed. The weight of the drained agent shall not exceed 3 percent of the applicable, full, agent-tank-capacity weight.

4.6.7.2 Discharge tests.- While discharging foam solution at maximum rated capacity, the maximum range of discharge for widely dispersed and solid stream foam shall be recorded for each type nozzle.

4.6.7.3 Proportioning system.- The proportioning system shall be tested as specified in MIL-T-4699, except the maximum discharge shall be that obtained through all water-foam nozzles, and the foam liquid percentage shall be measured for nominal rates of flow of 100, 300, 400, 500, 600, 800, 900, 1,000, 1,100, 1,300, 1,400 gpm.

4.6.7.4 Foam quality.- The foam quality shall be tested as specified in MIL-T-4699.

4.6.8 High temperature operation.- When equipped with the cab air-conditioner kit, the truck shall be filled with fuel and agent, instrumented, and placed in a chamber and subjected to a minimum ambient temperature of 125° F for at least 48 hours, or until temperature stabilization is reached. The air-conditioner kit shall be operated and shall obtain and maintain an ambient interior cab temperature of 90° F, or lower, within 15 minutes after starting.

4.6.9 Component endurance test.- Endurance tests need only to be conducted on those components which incorporate a different design and are substituted in lieu of those referred to on Drawing Index 531000. The endurance tests will represent, in the number of operations or duration of time of test, usage equivalent to that anticipated within a period of actual normal service life of not less than 5 years. These tests shall include, but not be limited to, operation of the applicable components for the number of cycles specified. A cycle shall consist of operating the component through all motions which the component would be subjected to during normal use.

Component	Number of Cycle
Cab doors	7,200
Cab windows	3,600

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Handline - open and close compartment door, reel and rewind hose	3,600
Turrets - complete operation of all component controls through full movement	3,600
Engagement of fire pump and pump drive	3,600
Seat adjusters	1,800
Compartment doors requiring daily use for maintenance, check of oil levels, et cetera	1,800
Fire pump primer	260
Removable panels	60
Hand brake	3,600
Hand throttle	260
Engine primer	1,800
Spotlight, complete operation	260
Windshield washer and wiper	520
Agent valve	1,800
Battery tray, rollout	260
Agent tank fill	900.

4.6.10 Reliability demonstration and test.- Satisfactory completion of all tests specified herein demonstrates compliance with the quantitative reliability requirements of this specification.

4.6.11 Maintainability verification.- Verification of the maintainability requirements shall be in accordance with MIL-M-26512, appendix B, paragraph entitled Maintenance Task Simulation.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

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6. NOTES

6.1 Intended use.- The A/S32P-2 truck is intended to be used for fire-fighting and rescue work involving aircraft and missiles where a high fire-hazard condition exists.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification

b. In the event it becomes necessary to change the drawing (see 3.2) in order to achieve the performance requirements of the specification, the contractor will immediately notify the contracting officer with recommendations for drawing changes to obtain the required performance. To the extent approved, such changes will be processed in accordance with the provisions of the changes clause of the contract

c. When the class II winterization kit is required (see 3.5.1)

d. When the air-conditioning kit is required (see 3.5.2)

e. The number of preproduction samples and the point of inspection (see 4.4.1)

f. Provisions for scheduling the electromagnetic interference test plan and report required by MIL-I-26600.

6.3 Definitions:

6.3.1 Reliability.- Reliability is the probability of performing without failure a specified function under given conditions for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence level: 90 percent

$$\text{MTBF} = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x} \quad \text{where } x = \frac{\text{Mission Time}}{\text{MTBF}}$$

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The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an exponential (2^{e} freedom) distribution even though the failure rate for the test is zero.

6.3.2 Mean-time-between-failures.- The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

6.3.3 Cross-country terrain.- Cross-country terrain consists of open fields, broken ground, loose sand or gravel road, and uneven terrain in general.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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